

EXP 25:

Perform Sharpening of Image using Gradient masking.

-1	-1	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

AIM: Sharpening of Image using Gradient masking.

Program:

```
import cv2

import numpy as np

img = cv2.imread(r'C:\Users\pavan\OneDrive\Opencv\x.jpg')

if img is None:

    print("Error: Image not found or path is incorrect")

else:

    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    gradient_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

    gradient_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

    gradient = cv2.magnitude(gradient_x, gradient_y)

    gradient = cv2.convertScaleAbs(gradient)

    cv2.imshow('Original Image', img)

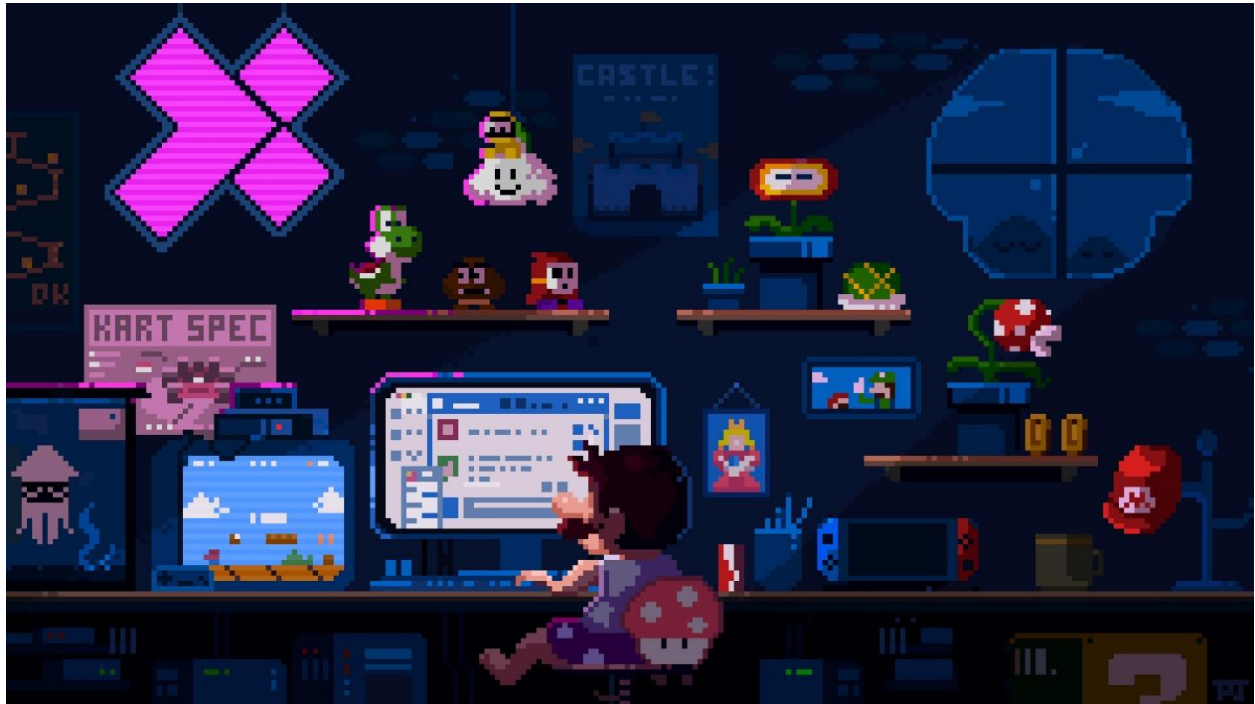
    cv2.imshow('Gradient Masked Sharpened Image', gradient)

    cv2.imwrite('Sharpened_Image_Exp25.jpg', gradient)

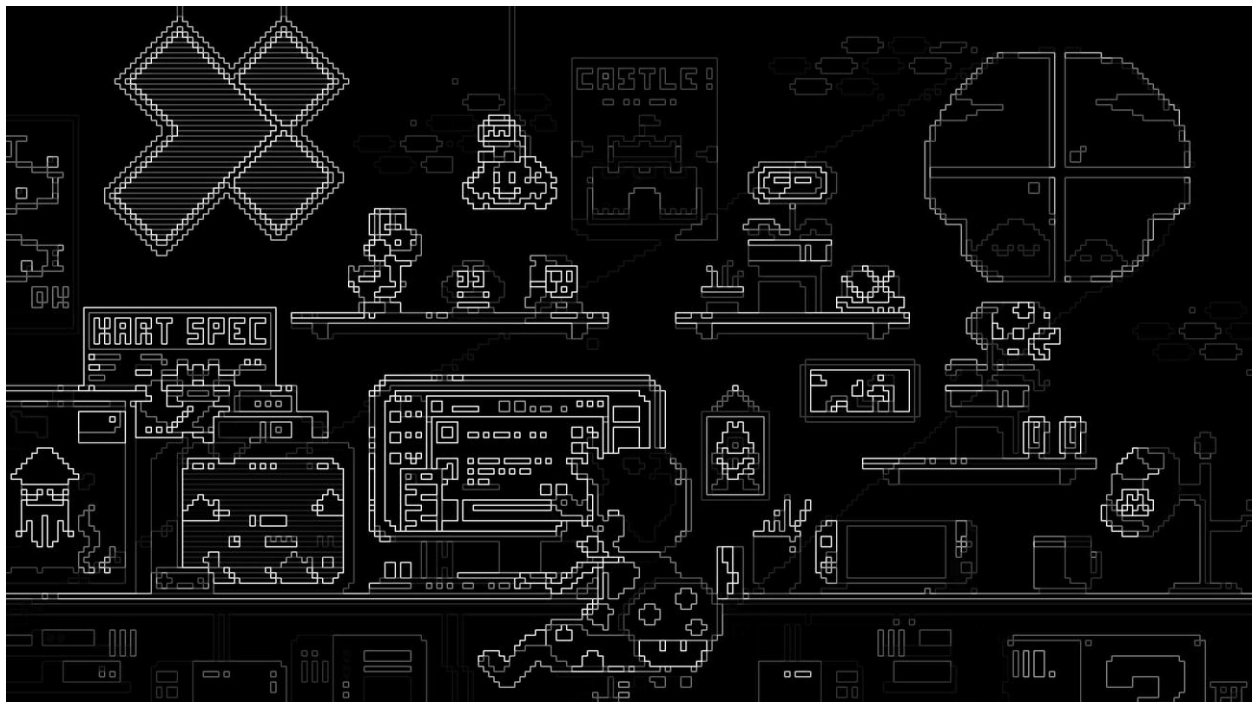
    cv2.waitKey(0)

    cv2.destroyAllWindows()
```

Input:



OUTPUT:



Result:

Thus the program is executed successfully.